

Total no. of Pages:

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NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

IInd SEMESTER, B. TECH (ECE / EIOT)

END-SEMESTER EXAMINATION, July 2023

Course Code: ECECC02/ EEECC02/ICECC02/ EIECC02

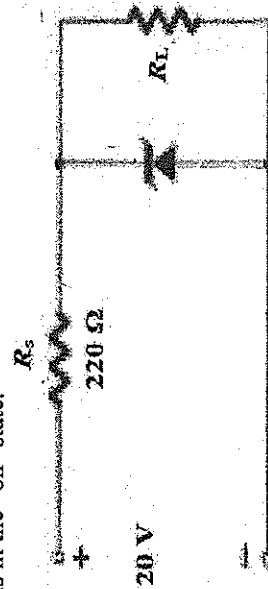
Course Title: Electronic Devices and Circuits

Time: 3:00 Hours

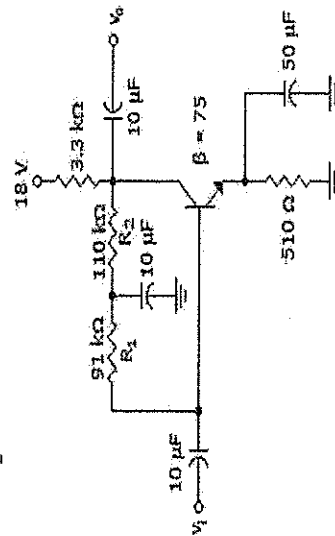
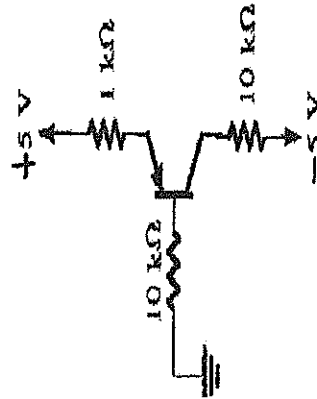
Max. Marks: 40

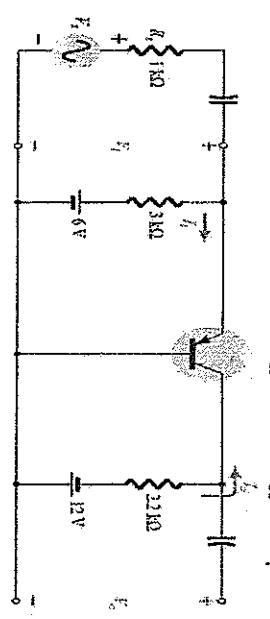
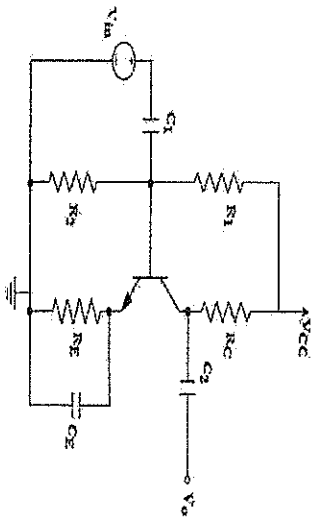
Note: Missing data/information if any, may be suitably assumed & mentioned in the answer.

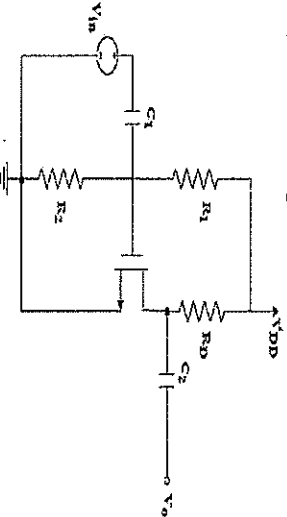
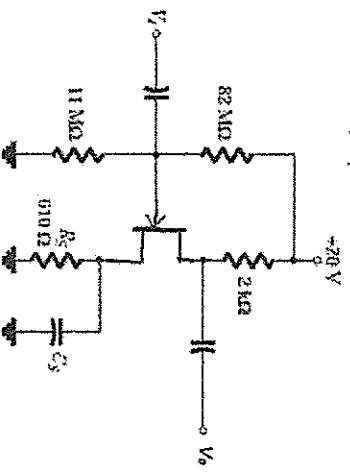
Q.No.	Question	Marks	CO
1	Attempt any two parts		
(a)	A pure semiconductor is doped with donor impurities to the extent of $1:10^5$ then, Find the resistivity and conductivity of the semiconductor. Also, determine, how many times the conductivity is increased in the semiconductor because of doping.	4	CO1
(b)	For the circuit shown below, the Zener voltage is 10 V and the maximum power dissipation is 400 mW. Determine: a. The value of R_L that will establish the maximum power conditions for the Zener diode. b. The minimum value of R_L to ensure that the Zener diode is in the "on" state.	4	CO1



(c)	Explain the following terms briefly: a. Diffusion Length b. Law of Mass Action c. Carrier Mobility d. Doping	4	CO1
2.	Attempt any two parts		
(a)	Explain the working of BJT in CE-configuration and explain its input, output and transfer characteristics?	4	CO2
(b)	Determine the voltages at all nodes and the currents through all branches for the given network. (Assume $\beta=50$)	4	CO2
(c)	For the circuit shown below, determine the value of I_{CQ} and V_{CEQ} .	4	CO2



3.	Attempt any two parts		
(a)	For the given common-base amplifier, determine the R_i, A_i, A_v and R_o using the complete hybrid equivalent model. $h_{ie} = 1.6 \text{ k}\Omega$ $h_{re} = 2 \times 10^{-4}$ $h_{fe} = 110$ $h_{oe} = 20 \text{ }\mu\text{S}$ 	4	CO3
(b)	Explain the construction and working of a simple current mirror circuit using BJTs.	4	CO3
(c)	For the circuit given below, determine the expression for (i) R_{in} (ii) R_{out} (iii) A_v 		CO3
4.	Attempt any two parts		
(a)	Explain the drain and transfer characteristics of n-channel enhancement and depletion type MOSFETs. Also, highlight the differences between JFET and BJT.	5	CO4

(b)	Determine the expression for the operating point (Q -point) of the circuit given below.	5	CO4
			
(c)	If $V_i = 20 \text{ mV}$, then determine the value of the input and output impedance for the following circuit, given that $I_{BSS} = 12 \text{ mA}$, $V_B = 3 \text{ V}$ and $r_d = 100 \text{ k}\Omega$. 	5	CO4
5.	Attempt any two parts		
(a)	Explain the basic principle and working of photodiodes	5	CO5
(b)	Describe various applications of Light Emitting Diodes with circuit diagrams.	5	CO5
(c)	Write short notes on: a. Phototransistor b. V-I characteristics of Uni-Junction Transistors	5	CO5